

HOTEL BOOKING

PERFORMANCE & CUSTOMER BEHAVIOR ANALYSIS

Excel • Power Query • SQL • Power BI

Hospitality / Business Intelligence / Customer Analytics

PREPARED BY

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TOOLS USED

Excel • SQL • Power BI
Power Query • DAX





Task 1 — Define Business Problem

Project overview and official task mapping



Data Period

2015–2017

PROBLEM STATEMENT

Analyze hotel booking data to understand booking trends, revenue patterns, cancellation drivers and customer behaviour. The final output should support decisions to improve revenue, reduce cancellations and optimize operations.

TASK 1 SOLUTION

Objective: build a clean analytical model, run SQL-based business analysis, and create two Power BI dashboards. KPIs used: Total Bookings, Total Revenue, Cancellation Rate, Average Stay, Lead Time and Customer Type behaviour.

OFFICIAL PROJECT TASKS

Task 1

Define Business Problem

Objectives, KPIs, stakeholders and timeline

Task 2

Data Cleaning

Missing values, duplicates, data types and base columns

Task 3

SQL Analysis

Basic, revenue, cancellation, customer and advanced segments

Task 4

Extract Business-Focused Dataset

High-value / high-risk booking subset

Task 5

Power BI Dashboard

Executive overview + filtered dataset dashboard

Final

Business Decision Analysis

Insights, recommendations and conclusion



Task 2 — Data Cleaning Input

Excel / Power Query preparation before SQL and Power BI



Data Period
2015–2017

WHAT TASK

Prepare hotel booking data for analysis and dashboard building.

WHAT USED

Excel and Power Query

WHAT WE DID

- checked missing values
- validated data types
- created derived columns
- prepared clean model

Power Query work area screenshot

The screenshot shows the Power Query Editor window with the following components:

- File Menu:** File, Home, Transform, Add Column, View, Tools, Help.
- Home Tab:** Close & Apply, New Source, Recent Sources, Enter Data, Data source settings, Manage Parameters, Export query results, Refresh Preview, Advanced Editor, Manage.
- Transform Tab:** Choose Columns, Remove Columns, Keep Rows, Remove Rows, Sort, Split Column, Group By, Data Type: Decimal Number, Use First Row as Headers, Replace Values, Merge Queries, Append Queries, Combine Files.
- Queries [2]:** Query Errors - 8/29/2026 9:31:24 AM [1], Errors in Booking_Clean, Other Queries [1], Booking_Clean.
- Table:** A table with columns: Row Number, booking_id, hotel, lead_time, arrival_date_year, arrival_date_month. The table contains 28 rows of data.
- Query Settings:** Name: Errors in Booking_Clean, All Properties, APPLIED STEPS: Source, Detected Type Mismatches, Added Index, Kept Errors, Reordered Columns.

Row Number	booking_id	hotel	lead_time	arrival_date_year	arrival_date_month
1	347	Resort Hotel	47	2015	July
2	348	Resort Hotel	115	2015	July
3	349	Resort Hotel	79	2015	July
4	350	Resort Hotel	91	2015	July
5	351	Resort Hotel	98	2015	July
6	352	Resort Hotel	109	2015	July
7	353	Resort Hotel	0	2015	July
8	354	Resort Hotel	98	2015	July
9	355	Resort Hotel	0	2015	July
10	356	Resort Hotel	9	2015	July
11	357	Resort Hotel	62	2015	July
12	358	Resort Hotel	45	2015	July
13	359	Resort Hotel	42	2015	July
14	360	Resort Hotel	54	2015	July
15	361	Resort Hotel	51	2015	July
16	362	Resort Hotel	54	2015	July
17	363	Resort Hotel	58	2015	July
18	364	Resort Hotel	0	2015	July
19	365	Resort Hotel	60	2015	July
20	366	Resort Hotel	321	2015	July
21	367	Resort Hotel	40	2015	July
22	368	Resort Hotel	51	2015	July
23	369	Resort Hotel	38	2015	July
24	370	Resort Hotel	75	2015	July
25	371	Resort Hotel	101	2015	July
26	372	Resort Hotel	42	2015	July
27	373	Resort Hotel	40	2015	July
28					



Task 2 — Clean Output & Prepared Dataset

Excel / Power Query output evidence



Data Period
2015–2017

Corrected data types

Field	Type	Null	Key	Default	Extra
booking_id	text	YES			
hotel	text	YES			
lead_time	int	YES			
arrival_date_year	int	YES			
arrival_date_month	text	YES			
arrival_date_week_number	int	YES			
arrival_date_day_of_month	int	YES			
arrival_date	text	YES			
stays_in_weekend_nights	int	YES			
stays_in_week_nights	int	YES			
total_nights	int	YES			
adr	double	YES			
revenue	double	YES			
booking_changes	int	YES			
days_in_waiting_list	int	YES			

Validation / duplicate check

A	B	C	D
booking_id	reserved_room_type	assigned_room_type	Count of Booking ID
1	C	C	1
2	C	C	1
3	A	C	1
4	A	A	1
5	A	A	1
6	A	A	1
7	C	C	1
8	C	C	1
9	A	A	1
10	D	D	1
11	E	E	1
12	D	D	1
13	D	E	1
14	G	G	1
15	E	E	1
16	D	E	1
17	E	E	1
18	A	E	1
19	A	G	1
20	G	G	1
21	F	F	1
22	A	A	1

SQL-ready output sample

booking_id	lead_time	deposit_type	is_canceled	revenue	risk_reason
12601	471	Non Refund	1	229	High Lead Time + Non-Refund
12602	471	Non Refund	1	229	High Lead Time + Non-Refund
12604	471	Non Refund	1	229	High Lead Time + Non-Refund
12606	471	Non Refund	1	229	High Lead Time + Non-Refund
12607	471	Non Refund	1	229	High Lead Time + Non-Refund
12608	471	Non Refund	1	229	High Lead Time + Non-Refund
12617	462	Non Refund	1	234	High Lead Time + Non-Refund
12658	462	Non Refund	1	234	High Lead Time + Non-Refund
12659	462	Non Refund	1	234	High Lead Time + Non-Refund
12661	462	Non Refund	1	234	High Lead Time + Non-Refund
12662	462	Non Refund	1	234	High Lead Time + Non-Refund
12664	462	Non Refund	1	234	High Lead Time + Non-Refund
12665	462	Non Refund	1	234	High Lead Time + Non-Refund
12666	462	Non Refund	1	234	High Lead Time + Non-Refund
12668	462	Non Refund	1	234	High Lead Time + Non-Refund
12669	462	Non Refund	1	234	High Lead Time + Non-Refund
12670	462	Non Refund	1	234	High Lead Time + Non-Refund
12671	462	Non Refund	1	234	High Lead Time + Non-Refund
12672	462	Non Refund	1	234	High Lead Time + Non-Refund
12673	462	Non Refund	1	234	High Lead Time + Non-Refund
12674	462	Non Refund	1	234	High Lead Time + Non-Refund
12676	462	Non Refund	1	234	High Lead Time + Non-Refund
12677	462	Non Refund	1	234	High Lead Time + Non-Refund
12678	462	Non Refund	1	234	High Lead Time + Non-Refund
12679	462	Non Refund	1	234	High Lead Time + Non-Refund
12680	462	Non Refund	1	234	High Lead Time + Non-Refund
12627	450	Non Refund	1	420.6	High Lead Time + Non-Refund

WHAT THE CLEAN OUTPUT SUPPORTS

Total Nights

weekend nights + weekday nights

Total Guests

adults + children + babies

Revenue

ADR × total nights

Arrival Month

formatted month field



Task 3 — SQL Analysis

Segment 1: Basic Understanding • Segment 2: Revenue Analysis



Data Period
2015–2017

PROBLEM STATEMENT

Use SQL queries to understand booking volume and revenue contribution across hotel, year and country.

SQL CONCEPTS USED

- SELECT
- COUNT / SUM
- GROUP BY
- ORDER BY / LIMIT
- JOIN

Segment 1 Query

```
1  -- 1. Total Number of Bookings
2
3  • SELECT COUNT(*) AS total_bookings
4    FROM booking;
5
6  -- 2. Number of Bookings by Hotel Type
7
8  • SELECT
9      hotel,
10     COUNT(*) AS total_bookings
11   FROM booking
12  GROUP BY hotel;
13
14  -- 3. Booking Distribution by Year
15
16  • SELECT
17      arrival_date_year,
18      COUNT(*) AS total_bookings
19   FROM booking
20  GROUP BY arrival_date_year
21  ORDER BY arrival_date_year;
22
23  -- 4. Top 10 Countries by Number of Bookings
24  • SELECT
25      country,
26      COUNT(*) AS total_bookings
27   FROM guest
28  WHERE country <> 'Unknown'
29  GROUP BY country
30  ORDER BY total_bookings DESC
31  LIMIT 10;
```

Segment 2 Query

```
1  -- 5. Total Revenue
2
3  • SELECT
4      ROUND(SUM(revenue), 2) AS total_revenue
5    FROM booking;
6
7  -- 6. Revenue by Hotel Type
8
9  • SELECT
10     hotel,
11     ROUND(SUM(revenue), 2) AS total_revenue
12   FROM booking
13  GROUP BY hotel;
14
15  -- 7. Monthly Revenue Trend
16
17  • SELECT
18      arrival_date_year,
19      arrival_month,
20      ROUND(SUM(revenue), 2) AS total_revenue
21   FROM booking
22  GROUP BY
23      arrival_date_year,
24      arrival_month
25  ORDER BY
26      arrival_date_year,
27      arrival_month,
28      arrival_date_year,
29      FIELD(
30          arrival_month,
31          'January', 'February', 'March', 'April',
32          'May', 'June', 'July', 'August',
33          'September', 'October', 'November', 'December'
34      );
35
36  -- 8. Top 5 Countries by Revenue
37
38  • SELECT
39      g.country,
40      ROUND(SUM(b.revenue), 2) AS total_revenue
41   FROM booking b
42  JOIN guest g
43    ON b.booking_id = g.booking_id
44  WHERE g.country <> 'Unknown'
45  GROUP BY g.country
46  ORDER BY total_revenue DESC
47  LIMIT 5;
48
```



Task 3 — Segments 1 & 2 Output & Finding

Basic Understanding + Revenue Analysis output evidence



Data Period

2015–2017

OUTPUT SCREENSHOTS — 8 RESULTS

1. Total bookings

Result Grid	Filter Rows
total_bookings	
30000	

2. Count by hotel type

Result Grid	Filter Rows
hotel	total_bookings
Resort Hotel	30000

3. Distribution by year

arrival_date_year	total_bookings
2015	8314
2016	16920
2017	4766

4. Top countries by bookings

country	total_bookings
PRT	14559
GBR	4572
ESP	2949
IRL	1573
FRA	996
DEU	840
CH	404
NLD	349
USA	305
CHE	305

RESULT

30,000 bookings were validated. Resort Hotel is the only hotel type. Revenue output is strongest from Portugal, Great Britain and Spain.

5. Total revenue

total_revenue
13124913.850000054

6. Revenue by hotel type

hotel	total_revenue
Resort Hotel	13124913.850000054

7. Monthly revenue trend

arrival_date_year	total_revenue
2015 Jan	444071.1
2015 Feb	557675.9
2015 Mar	1000000.0
2015 Apr	1000000.0
2015 May	1000000.0
2015 Jun	1000000.0
2015 Jul	1000000.0
2015 Aug	1000000.0
2015 Sep	1000000.0
2015 Oct	1000000.0
2015 Nov	1000000.0
2015 Dec	1000000.0
2016 Jan	1000000.0
2016 Feb	1000000.0
2016 Mar	1000000.0
2016 Apr	1000000.0
2016 May	1000000.0
2016 Jun	1000000.0
2016 Jul	1000000.0
2016 Aug	1000000.0
2016 Sep	1000000.0
2016 Oct	1000000.0
2016 Nov	1000000.0
2016 Dec	1000000.0
2017 Jan	1000000.0
2017 Feb	1000000.0
2017 Mar	1000000.0
2017 Apr	1000000.0
2017 May	1000000.0
2017 Jun	1000000.0
2017 Jul	1000000.0
2017 Aug	1000000.0

8. Top countries by revenue

country	total_revenue
PRT	5775975.160000003
GBR	2239575.6099999915
ESP	1333734.080000001
IRL	876528.5500000007
FRA	454351.27999999974

INSIGHT

Revenue is concentrated in a few countries, so country-level review is useful for market planning.

TAKEAWAY

Use SQL outputs as evidence before converting the same logic into Power BI visuals.



Task 3 — SQL Analysis

Segment 3: Cancellation Analysis • Segment 4: Customer Behavior



Data Period
2015–2017

PROBLEM STATEMENT

Use SQL to identify cancellation drivers and understand customer type behaviour.

SQL CONCEPTS USED

- CASE WHEN
- AVG
- SUM
- GROUP BY
- ORDER BY

Segment 3 Query Screenshot

```
1 -- 9. Overall cancellation rate
2 SELECT COUNT(*) AS total_bookings,
3        SUM(is_canceled) AS cancelled_bookings,
4        ROUND(SUM(is_canceled)/COUNT(*)*100,2) AS cancellation_rate
5 FROM booking;
6
7 -- 10. Cancellation rate by hotel type
8 SELECT hotel, COUNT(*) AS total_bookings,
9        SUM(is_canceled) AS cancelled_bookings,
10       ROUND(SUM(is_canceled)/COUNT(*)*100,2) AS cancellation_rate
11 FROM booking
12 GROUP BY hotel;
13
14 -- 11. Cancellation vs deposit type
15 SELECT deposit_type, COUNT(*) AS total_bookings,
16        SUM(is_canceled) AS cancelled_bookings,
17        ROUND(SUM(is_canceled)/COUNT(*)*100,2) AS cancellation_rate
18 FROM booking
19 GROUP BY deposit_type
20 ORDER BY cancellation_rate DESC;
21
22 -- 12. Avg lead time for cancelled vs non-cancelled
23 SELECT CASE WHEN is_canceled=1 THEN 'Cancelled' ELSE 'Not Cancelled' END AS booking_status,
24        COUNT(*) AS total_bookings,
25        ROUND(AVG(lead_time),2) AS average_lead_time
26 FROM booking
27 GROUP BY booking_status;
```

Segment 4 Query Screenshot

```
1 -- 13. Repeat vs new guests count
2 SELECT CASE WHEN is_repeated_guest=1 THEN 'Repeat Guest' ELSE 'New Guest' END AS guest_type,
3        COUNT(*) AS total_bookings
4 FROM booking
5 GROUP BY guest_type;
6
7 -- 14. Revenue by customer type
8 SELECT customer_type,
9        COUNT(*) AS total_bookings,
10       ROUND(SUM(revenue),2) AS total_revenue
11 FROM booking
12 GROUP BY customer_type
13 ORDER BY total_revenue DESC;
14
15 -- 15. Avg stay duration by segment
16 SELECT market_segment,
17        COUNT(*) AS total_bookings,
18        ROUND(AVG(total_nights),2) AS average_stay_nights
19 FROM booking
20 GROUP BY market_segment;
21
22 -- 16. Special requests vs booking success
23 SELECT total_of_special_requests,
24        COUNT(*) AS total_bookings,
25        SUM(CASE WHEN is_canceled=0 THEN 1 ELSE 0 END) AS successful_bookings,
26        ROUND(SUM(CASE WHEN is_canceled=0 THEN 1 ELSE 0 END)/COUNT(*)*100,2) AS success_rate
27 FROM booking
28 GROUP BY total_of_special_requests;
```



Task 3 — Segments 3 & 4 Output & Finding

Cancellation Analysis + Customer Behavior output evidence



Data Period

2015–2017

OUTPUT SCREENSHOTS — 8 RESULTS

9. Overall cancellation

	total_bookings	cancelled_bookings	cancellation_rate
▶	30000	11114	37.05

10. Cancellation by hotel

hotel	total_bookings	cancelled_bookings	cancellation_rate
▶ Resort Hotel	30000	11114	37.05

11. Deposit type risk

deposit_type	total_bookings	cancelled_bookings	cancellation_rate
▶ Non Refund	1711	1690	98.43
No Deposit	28148	9442	33.54
Refundable	141	22	15.60

12. Lead time comparison

booking_status	total_bookings	average_lead_time
▶ Cancelled	11114	128.74
Not Cancelled	18886	78.64

RESULT

Non-Refund has the highest cancellation rate. Cancelled bookings also show higher average lead time.

13. Repeat vs new guest

guest_type	total_bookings
▶ New Guest	28533
Repeat Guest	1467

14. Revenue by customer type

customer_type	total_bookings	total_revenue
▶ Transient	23096	10682246.76
Transient-Party	5467	15942777.41
Contract	1224	803264.03
Group	213	45125.65

15. Avg stay by segment

market_segment	total_bookings	average_stay_nights
▶ Offline TA/TO	5673	6.01
Online TA	13069	4.33
Groups	4672	3.93
Direct	4574	3.61
Corporate	1882	2.38
Complementary	130	2.18

16. Special requests success

total_of_special_requests	total_bookings	successful_bookings	booking_success_rate
▶ 0	17729	1018	5.73
1	8312	5717	68.78
2	2303	2177	94.53
3	570	404	70.88
4	26	61	232.30
5	10	9	90.00

INSIGHT

Deposit type and lead time explain cancellation risk better than booking count alone.

TAKEAWAY

Customer and cancellation outputs support the second Power BI dashboard.



Task 3 — SQL Analysis

Segment 5: Advanced Analysis • Segment 6: Additional Advanced Queries



Data Period
2015–2017

PROBLEM STATEMENT

Use advanced SQL to rank revenue, track running totals, identify high-risk bookings and study revenue contribution.

SQL CONCEPTS USED

- Window functions
- RANK / LAG
- Running total
- Contribution %
- Factor combinations

Segment 5 Query Screenshot

```
1 -- 17. Rank top months by revenue
2 SELECT arrival_date_year, arrival_month,
3         ROUND(SUM(revenue),2) AS total_revenue,
4         RANK() OVER (ORDER BY SUM(revenue) DESC) AS revenue_rank
5 FROM booking
6 GROUP BY arrival_date_year, arrival_month;
7
8 -- 18. Identify high-risk bookings
9 SELECT booking_id, lead_time, deposit_type, market_segment,
10        is_canceled, revenue
11 FROM booking
12 WHERE lead_time > 180 OR deposit_type = 'Non Refund'
13 ORDER BY lead_time DESC;
14
15 -- 19. Running total of bookings over time
16 SELECT proper_arrival_date,
17        COUNT(*) AS daily_bookings,
18        SUM(COUNT(*)) OVER (ORDER BY proper_arrival_date) AS running_total_bookings
19 FROM booking
20 GROUP BY proper_arrival_date;
21
22 -- 20. Segment with high revenue but high cancellations
23 SELECT market_segment, COUNT(*) AS total_bookings,
24        ROUND(SUM(revenue),2) AS total_revenue,
25        SUM(is_canceled) AS cancelled_bookings,
26        ROUND(SUM(is_canceled)/COUNT(*)*100,2) AS cancellation_rate
27 FROM booking
28 GROUP BY market_segment;
```

Segment 6 Query Screenshot

```
1 -- 21. Revenue contribution by customer segment
2 SELECT customer_type, COUNT(*) AS total_bookings,
3        ROUND(SUM(revenue),2) AS total_revenue,
4        ROUND(SUM(revenue) / SUM(SUM(revenue)) OVER () * 100,2) AS revenue_contribution_pct
5 FROM booking
6 GROUP BY customer_type;
7
8 -- 22. Rank customers within each segment by revenue
9 SELECT customer_type, country, COUNT(*) AS total_bookings,
10        ROUND(SUM(revenue),2) AS total_revenue,
11        RANK() OVER (PARTITION BY customer_type ORDER BY SUM(revenue) DESC) AS revenue_rank
12 FROM booking
13 GROUP BY customer_type, country;
14
15 -- 23. Top 3 countries by cancellation rate
16 SELECT country, total_bookings, cancelled_bookings, cancellation_rate
17 FROM country_cancellation_summary
18 ORDER BY cancellation_rate DESC
19 LIMIT 3;
20
21 -- 24-25. Repeat trend and highest cancellation factor combinations
22 SELECT arrival_date_year, repeat_bookings, repeat_booking_growth_pct FROM repeat_growth;
23 SELECT lead_time_group, deposit_type, market_segment, cancellation_rate FROM risk_combo;
```



Task 3 — Segments 5 & 6 Output & Finding

Advanced SQL output evidence



Data Period
2015–2017

OUTPUT SCREENSHOTS — 9 RESULTS

17. Revenue rank by month

	arrival_date_year	arrival_month	total_revenue	revenue_rank
▶	2016	August	1598727.45	1
	2016	July	1179067.98	2
	2015	August	1176179.48	3
	2015	July	885263.67	4
	2017	August	859978.19	5
	2017	July	777975.36	6
	2016	September	750103.34	7
	2016	June	720613.8	8
	2015	September	638048.22	9
	2016	May	535567.9	10

18. High-risk bookings

lead_time_group	deposit_type	market_segment	total_bookings	cancelled_bookings	cancellation_rate
Lower Lead Time	No Deposit	Online TA	9533	3762	39.54
High Lead Time	No Deposit	Online TA	3539	2451	69.65
High Lead Time	Non Refund	Group	988	957	96.89
Lower Lead Time	No Deposit	Direct	2623	651	24.80
High Lead Time	No Deposit	Offline TA/TO	2425	553	22.80
High Lead Time	No Deposit	Group	1779	510	28.67
Lower Lead Time	Non Refund	Go [omitted]	957	497	51.93
Lower Lead Time	No Deposit	Group	1280	497	38.84
Lower Lead Time	No Deposit	Offline TA/TO	3095	423	13.69
Lower Lead Time	No Deposit	Corporate	8834	331	3.75
High Lead Time	No Deposit	Direct	647	221	34.16
High Lead Time	Non Refund	Offline TA/TO	138	136	98.55
Lower Lead Time	No Deposit	Complementary	127	32	25.20
Lower Lead Time	Non Refund	Online TA	38	38	100.00
Lower Lead Time	Non Refund	Corporate	25	38	72.00

19. Running total

proper_arrival_date	daily_bookings	running_total_bookings
▶	2015-07-13	12024
	2015-07-14	12075
	2015-07-15	12126
	2015-07-16	12177
	2015-07-17	12228
	2015-07-18	12279
	2015-07-19	12330
	2015-07-20	12381
	2015-07-21	12432
	2015-07-22	12483
	2015-07-23	12534
	2015-07-24	12585
	2015-07-25	12636
	2015-07-26	12687

RESULT

Advanced SQL outputs compared top revenue months, running booking trend, cancellation-heavy segments and repeat-customer growth.

20. High revenue + cancellations

market_segment	total_bookings	total_revenue	cancelled_bookings	cancellation_rate
▶	Online TA	13069	6879357.93	47.81
	Offline TA/TO	5673	2692865.96	20.04
	Direct	4574	2121951.29	8.73
	Groups	4672	1192172.47	24.74
	Corporate	3882	238034.7	18.54
	Complementary	130	527.5	33

21. Revenue contribution %

	customer_type	total_bookings	total_revenue	revenue_contribution_pct
▶	Transient	23096	10682246.76	81.39
	Transient-Party	5467	1594277.41	12.15
	Contract	1224	803264.03	6.12
	Group	213	45125.65	0.34

22. Customer ranking

customer_type	country	total_bookings	total_revenue	revenue_pct
Contract	GBR	800	53056.8	1
Contract	PER	242	151817.04	2
Contract	IND	82	48676.35	3
Contract	DEU	24	20534.38	4
Contract	NLD	14	11234.43	5
Contract	ESP	18	9424.8	6
Contract	POL	7	8629.23	7
Contract	CHE	4	4620.98	8
Contract	NOR	4	3677.92	9
Contract	FIN	4	3239.08	10
Contract	OMN	3	2789.8	11
Contract	BRA	4	2682.95	12
Contract	FRA	4	2465.22	13
Contract	ITA	3	2433.93	14
Contract	SRB	2	2133.25	15

INSIGHT

Advanced queries move the work from simple reporting to management-focused decision analysis.

23. Top cancellation countries

country	total_bookings	cancelled_bookings	cancellation_rate
▶	GEO	7	100.00
	FJI	1	100.00
	GGY	1	100.00

24. Repeat growth trend

	arrival_date_year	total_bookings	repeat_bookings	repeat_bookings_share	previous_year_repeat_bookings	repeat_bookings_growth_pct
▶	2015	8314	134	1.61	688	5.05
	2016	16920	837	4.95	124	524.63
	2017	4768	486	10.41	837	-40.74

25. Risk factor combination

lead_time_group	deposit_type	market_segment	total_bookings	cancelled_bookings	cancellation_rate
▶ Lower Lead Time	No Deposit	Online TA	9533	3762	39.54
High Lead Time	No Deposit	Online TA	3539	2451	69.65
High Lead Time	Non Refund	Groups	988	957	96.89
Lower Lead Time	No Deposit	Direct	2623	651	24.80
High Lead Time	No Deposit	Offline TA/TO	2425	553	22.80
High Lead Time	No Deposit	Groups	1779	510	28.67
Lower Lead Time	Non Refund	Go 	957	497	51.93
Lower Lead Time	No Deposit	Groups	1280	497	38.84
Lower Lead Time	No Deposit	Offline TA/TO	3095	423	13.69
Lower Lead Time	No Deposit	Corporate	8834	331	3.75
High Lead Time	No Deposit	Direct	647	221	34.16
High Lead Time	Non Refund	Offline TA/TO	138	136	98.55
Lower Lead Time	No Deposit	Complementary	127	32	25.20
Lower Lead Time	Non Refund	Online TA	38	38	100.00
Lower Lead Time	Non Refund	Corporate	25	38	72.00

TAKEAWAY

These outputs justify creating a focused Priority_Bookings dataset for deeper dashboard analysis.



Task 4 — Extract Business-Focused Dataset



Data Period
2015–2017

WHAT IS PRIORITY_BOOKINGS?

Priority_Bookings is a business-focused subset extracted from the cleaned booking data. It keeps bookings that need closer management attention because they are high-risk, high-value or operationally important.

FILTERING LOGIC

Recent Years

Focused on recent booking behaviour

Target Customer Types

Transient and Transient-Party customer segments

Risk Signals

Cancelled OR high lead time OR no deposit

Decision Use

Supports risk review, revenue protection and cancellation reduction

One-line PPT explanation: Priority_Bookings is the filtered high-risk / high-value booking dataset used for deeper management analysis.

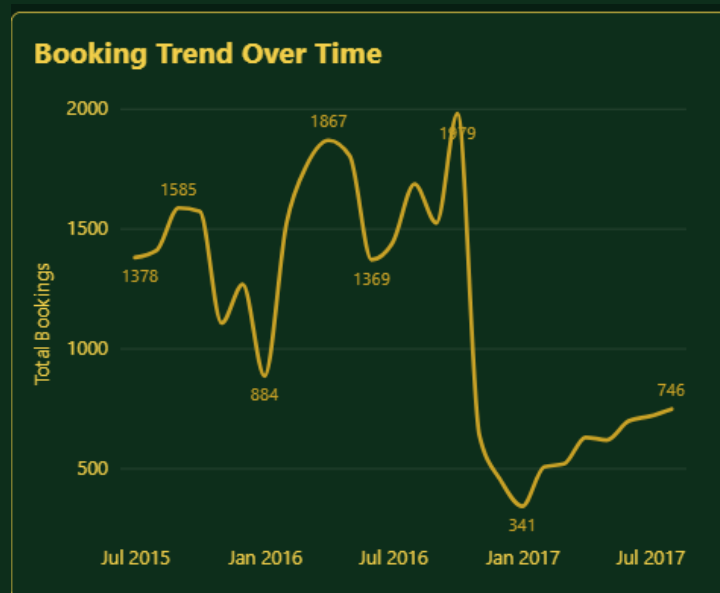


Task 5 — Power BI Dashboard 1: Executive Overview

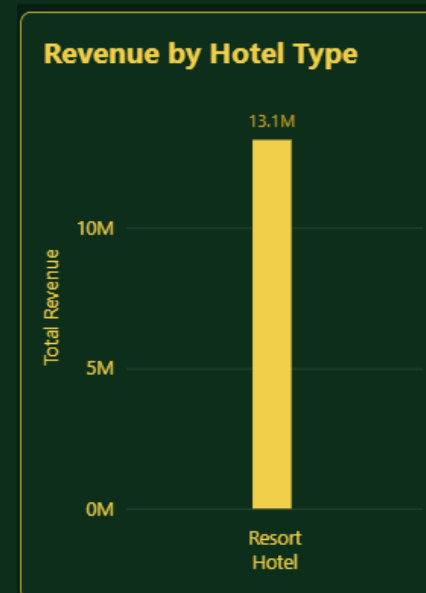


Data Period
2015–2017

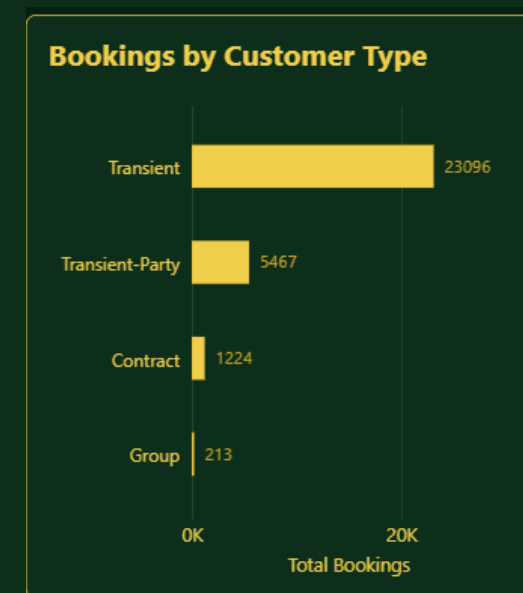
Booking Trend Over Time



Revenue by Hotel Type



Bookings by Customer Type





Task 5 — Dashboard 1: KPI Cards & Filters



Data Period
2015–2017

KPI Cards

Total Bookings
30000

Total Revenue
13.12M

Cancellation Rate
37.05%

WHAT TASK

Display main business performance before detailed analysis.

WHAT USED

KPI cards and slicers using full-data Power BI model.

WHAT WE GOT

30,000 bookings, ₹13.12M revenue and 37.05% cancellation rate.

Filters

Filters

Year

- ☐ 2015
- ☐ 2016
- ☐ 2017

Customer Type

- ☐ Contract
- ☐ Group
- ☐ Transient
- ☐ Transient-Party

Hotel Type

- ☒ Resort Hotel



Task 5 — Cancellation Insights



Data Period
2015–2017

WHAT TASK

Analyze cancellation risk by lead time, deposit type, market segment and risk category.

WHAT USED

Column and bar charts

WHAT WE GOT

Segment visuals are arranged for management review.

Power BI output



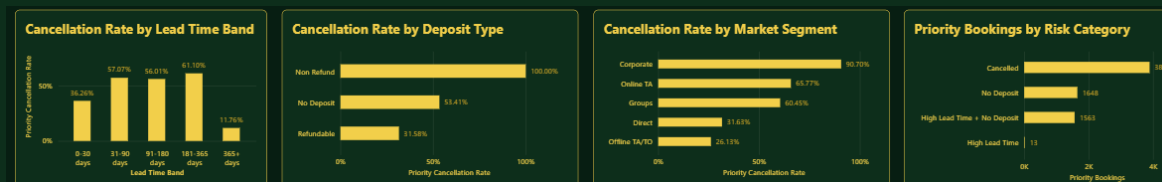


Cancellation Insights — Output & Finding



Data Period
2015–2017

Output Screenshot



RESULT

Priority cancellation rate is 54.67% with 3,889 cancelled priority bookings.

INSIGHT

Lead time and deposit type are key cancellation risk dimensions.

TAKEAWAY

Monitor priority-risk groups before arrival date to reduce avoidable cancellations.



Task 5 — Revenue Insights



Data Period
2015–2017

WHAT TASK

Study where priority revenue comes from and which drivers explain revenue contribution.

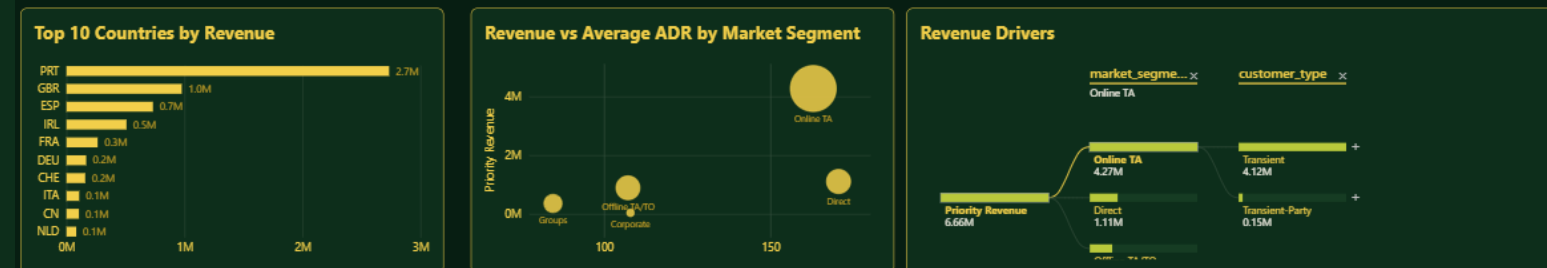
WHAT USED

Bar chart, scatter plot and decomposition tree

WHAT WE GOT

Revenue contribution can be reviewed by country, ADR and driver path.

Power BI output





Revenue Insights — Output & Finding



Data Period
2015–2017

Revenue Row



RESULT

Portugal is the top priority revenue country; revenue drivers show Online TA and Transient customers as key contributors.

INSIGHT

Revenue should be compared through contribution, not booking count alone.

TAKEAWAY

Use decomposition tree to find where revenue comes from before planning actions.

Revenue Drivers





Task 5 — Customer Insights



Data Period
2015–2017

WHAT TASK

Understand repeat guests, special requests and customer type behavior within Priority_Bookings.

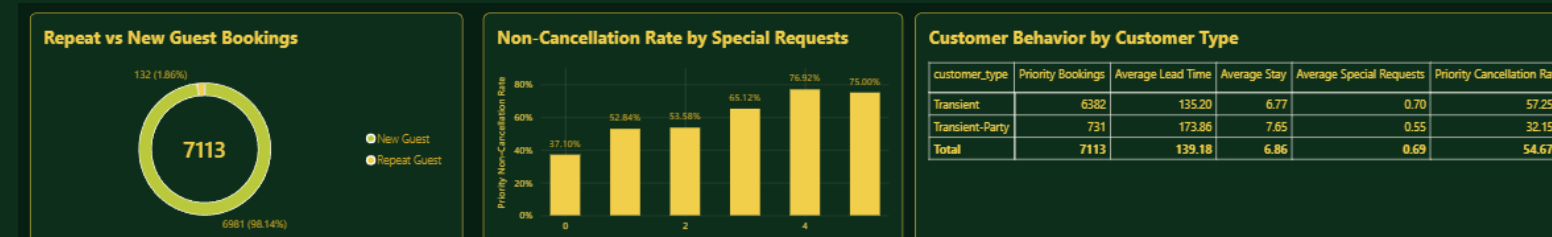
WHAT USED

Donut chart, column chart and matrix

WHAT WE GOT

Customer behavior metrics are grouped for management review.

Power BI output





Segment 3 — Customer Insights



Data Period
2015–2017

Customer Row



RESULT

New guests dominate Priority_Bookings, while special requests improve non-cancellation rate.

INSIGHT

Customer behaviour is useful for service planning and retention strategy.

TAKEAWAY

Encourage useful pre-arrival requests and track customer type behaviour regularly.

Customer Behavior Matrix

Customer Behavior by Customer Type

customer_type	Priority Bookings	Average Lead Time	Average Stay	Average Special Requests	Priority Cancellation Rate
Transient	6382	135.20	6.77	0.70	57.25%
Transient-Party	731	173.86	7.65	0.55	32.15%
Total	7113	139.18	6.86	0.69	54.67%



Final Dashboard 1 — Executive Overview

Full-dataset Power BI dashboard



Data Period
2015–2017



Hotel Booking Performance Dashboard

Bookings, Revenue & Customer Insights



Data Period
2015–2017

Total Bookings

30000

Total Revenue

13.12M

Cancellation Rate

37.05%

Filters

Year

☐ 2015

☐ 2016

☐ 2017

Customer Type

☐ Contract

☐ Group

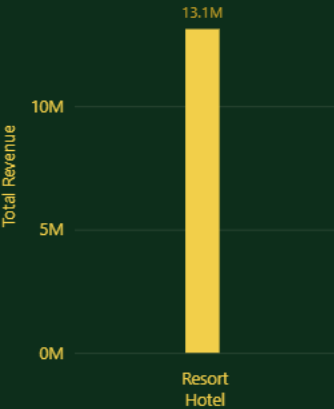
☐ Transient

☐ Transient-Party

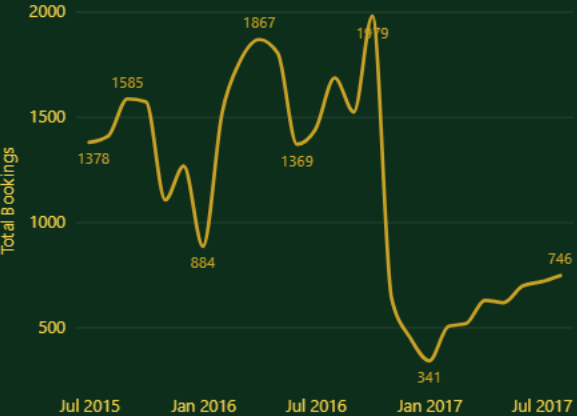
Hotel Type

☒ Resort Hotel

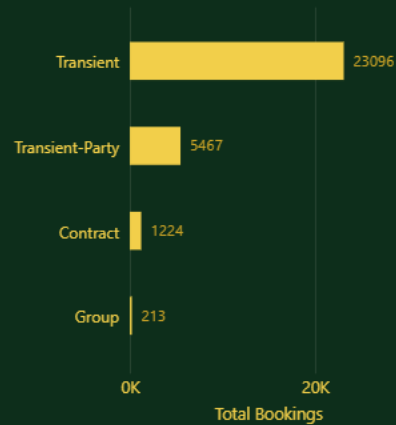
Revenue by Hotel Type



Booking Trend Over Time



Bookings by Customer Type





Final Dashboard 2 — Priority Analysis

Filtered / derived dataset dashboard



Data Period
2015–2017



Priority Booking Risk & Revenue Analysis

Cancellation, Revenue & Customer Behaviour Insights



Data Period
2015–2017

Priority Bookings

7,113

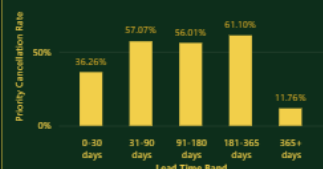
Priority Cancelled Bookings

3,889

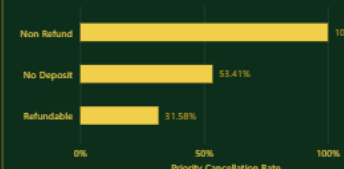
Priority Cancellation Rate

54.67%

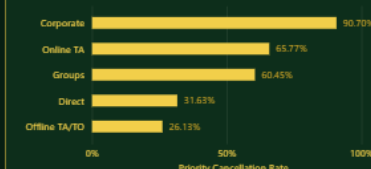
Cancellation Rate by Lead Time Band



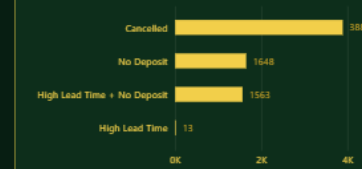
Cancellation Rate by Deposit Type



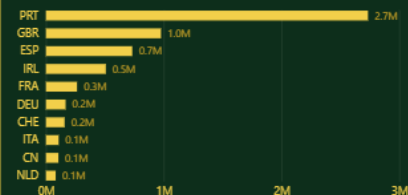
Cancellation Rate by Market Segment



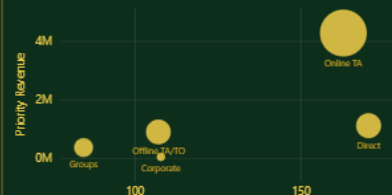
Priority Bookings by Risk Category



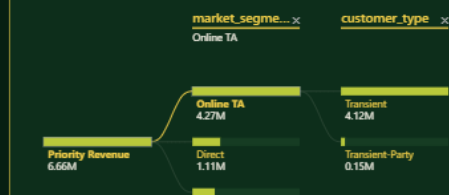
Top 10 Countries by Revenue



Revenue vs Average ADR by Market Segment



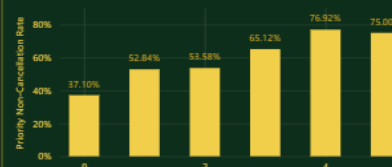
Revenue Drivers



Repeat vs New Guest Bookings



Non-Cancellation Rate by Special Requests



Customer Behavior by Customer Type

customer_type	Priority Bookings	Average Lead Time	Average Stay	Average Special Requests	Priority Cancellation Rate
Transient	6382	135.20	6.77	0.70	57.25%
Transient-Party	731	173.86	7.65	0.55	32.15%
Total	7113	139.18	6.86	0.69	54.67%



Business Decision Analysis



Data Period
2015–2017

Booking Scope

The available dataset contains 30,000 Resort Hotel bookings.

Revenue Pattern

Portugal, Great Britain and Spain are the strongest revenue markets.

Cancellation Risk

Non-Refund and long-lead bookings show higher cancellation concern.

Customer Behavior

Transient customers dominate booking volume and revenue contribution.

Priority Analysis

Priority_Bookings helps focus on higher-risk / high-value bookings.

Dashboard Use

Two dashboards separate executive review from deeper risk and revenue analysis.



Business Recommendations

Actions based on project findings



Data Period

2015–2017

1

Reduce High-Risk Cancellations

Monitor long lead-time and no-deposit bookings before arrival using priority alerts.

2

Protect Key Revenue Markets

Track Portugal, Great Britain and Spain separately for revenue and cancellation movement.

3

Review Deposit Strategy

Use stricter follow-up for high-risk deposit groups instead of treating all bookings equally.

4

Use Customer-Type Strategy

Create communication and retention plans for Transient and Transient-Party customers.

5

Monitor Special Requests

Encourage special requests because they are linked with better booking success.

6

Use Dashboards for Reviews

Use full dataset dashboard for management review and Priority dashboard for risk action.

THANK YOU

HOTEL BOOKING PERFORMANCE & CUSTOMER BEHAVIOR ANALYSIS

PREPARED BY

D. Jaya Kumar

